

Autonomous Area Scanning and Precision Landing on a Visual Target with a Parrot Mambo Minidrone

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Abstract—This report documents a search-and-land mission: a Parrot Mambo minidrone sweeps an indoor arena in a scanning flight pattern, detects a colored landing pad below using its downward-facing camera, and transitions from search to a centered vertical descent onto the pad. The model derives from the MathWorks Minidrone Competition project structure, with the author's scanning references, color-detection thresholds, and landing handover logic. Detection uses color segmentation and blob centroid extraction on the live camera feed. The deployed flight finds the pad and lands on it autonomously in one continuous take.

Index Terms—Parrot Mambo, Simulink, search and land, color segmentation, blob analysis, precision landing, Minidrone Competition.

I. INTRODUCTION

Line following assumes the path is already beneath the drone. This mission removes that assumption: the target's location is not given, so the drone must first search the arena, then recognize the landing pad, then land on it, chaining three behaviors into one autonomous flight.

The Parrot Mambo carries the sensor suite the whole series relies on: an ultrasound sensor timing pulses to the floor for altitude, a pressure sensor for altitude changes, an IMU providing linear acceleration and angular rates, and a downward-facing camera whose optical flow contributes horizontal velocity estimates. Four motors in an X configuration are commanded through motor mixing: collective thrust, a differential between the clockwise and counter-clockwise pairs for yaw, and side-to-side speed imbalances for pitch and roll. The airframe is underactuated, with six degrees of freedom and only four inputs, so every lateral motion begins as a deliberate tilt of the thrust vector.

II. SYSTEM ARCHITECTURE

All flight code was developed in Simulink against the Parrot Minidrone Support Package and deployed to the physical airframe, keeping the model structure the support package expects: an estimator consuming the sensor buses, a flight control system, and reference logic that the author modified per task. Structuring the work as small reusable subsystems made iteration and debugging on hardware considerably less painful.

The model is built on the Minidrone Competition project structure, which supplies the airframe, estimator, and flight control scaffolding; the author's contribution lives in the scanning flight references, the vision thresholds tuned to the pad's signature color, and the logic that arbitrates between search, centering, and descent.

III. METHOD

A. Search

A scanning pattern carries the drone across the arena at constant altitude while every frame from the downward camera is thresholded against the pad's color signature.

B. Detection and centering

When the pad enters the field of view, blob analysis extracts the centroid of the detection, and the offset from the image center drives lateral corrections that bring the drone over the target.

C. Landing

Once centered, control hands over to the landing sequence and the drone descends vertically onto the pad.

IV. FLIGHT TESTING AND RESULTS

In the deployed flight over the checkerboard arena, the drone scans, acquires the colored pad, centers itself, and lands on the target autonomously, completing the full search-detect-land chain in a single continuous flight.

V. CHALLENGES

- Detecting a small pad from a moving, vibrating camera without false positives from the arena's own pattern.
- Transitioning cleanly from scanning motion to a centered descent without overflying the target.
- Color thresholds holding up across the arena's lighting.

VI. CONCLUSION

The mission demonstrates behavior arbitration on top of the series' vision and control stack: search, recognition, and precision landing chained autonomously on real hardware.

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